Lightweight Transformer and CNN Models Comparison Analysis for Degenerative Lumbar Spine Classification

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***Abstract*—This electronic document is a "live" template and already defines the components of your paper [title, text, heads, etc.] in its style sheet. **CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract*. (*Abstract*)**

Keywords—Transformer, Convolutional Neural Network (CNN), Degenerative Lumbar Spine, Inception-V4, Vision Transformer (ViT), EfficientNet, EfficientNetV2

I. INTRODUCTION

Low back pain is the leading cause of years lived with disability (YLDs) worldwide and is becoming a significant issue in an aging population. The primary contributors to mechanical lumbar spine disorders for which are responsible for up to 97% of low back pain diagnoses include degenerative intervertebral disc issues, lumbar spondylotic radiculopathy, spondylolisthesis, and spinal stenosis, all of which are part of a degenerative process affecting the lumbar spine [1].

A significant challenge in managing low back pain is that approximately 85 percent of patients with isolated low back pain do not receive a precise pathoanatomical diagnosis [2]. This lack of clarity complicates treatment decisions, as the relationship between symptoms reported by patients and the findings from imaging studies is often weak. Many individuals may present with similar symptoms, yet their imaging results can reveal diverse and sometimes ambiguous findings [3].

Therefore, there will be a need for an accurate classification of the degenerative lumbar spine condition. It is essential to ensure timely diagnosis and implement effective treatment strategies. By correctly identifying the severity of the condition, healthcare providers can prioritize which patients require more careful evaluation and intervention, enhancing patient outcomes. This targeted approach ensures that those with more severe conditions receive timely and appropriate care, while also optimizing resource allocation within the healthcare system.

Alongside the rapid development of technology, artificial intelligence (AI), particularly deep learning, has become an increasingly vital tool in healthcare, especially for medical image processing which include computer vision, pattern recognition, image mining, and machine learning in several directions. Deep learning techniques, especially convolutional neural networks (CNNs), have demonstrated exceptional capabilities in analyzing and interpreting complex medical images such as MRI and CT scans. These models can automatically extract relevant features and patterns from imaging data [4]. Thus, many computer-aided detection and diagnosis (CAD) schemes have been developed and tested to help clinicians read medical images more efficiently and make more accurate and objective diagnostic decisions [5].

Moreover, the emergence of transformer-based models has significantly transformed the field of image analysis. Initially developed for natural language processing (NLP), these models have demonstrated their effectiveness as a compelling alternative to CNNs in computer vision, achieving performance levels comparable to those of CNNs [6]. Transformers leverage a powerful mechanism called attention, which allows them to focus on specific parts of the input for enhanced results. This approach has led to the exploration of various strategies for applying transformers to images, including splitting images into patches and treating them as tokens, which has resulted in competitive performance on tasks like image classification [7].

Addressing the urgency of an accurate classification of the degenerative lumbar spine condition given the significant disability burden of low back pain worldwide, this research aims to identify the optimal model for diagnosing these conditions. By evaluating and comparing the performance of CNN-based models, such as Inception-V4, EfficientNet, and EfficientNetV2, against lightweight transformer models like Vision Transformers (ViTs), this study seeks to determine which approach provides the highest accuracy and efficiency in classifying lumbar spine conditions as normal, moderate, or severe. As well as offering valuable insights into the strengths and limitations of each model type, guiding the selection of an effective tool for spine diagnostics.

This research stands apart by directly comparing lightweight transformer models, specifically Vision Transformers (ViTs), with state-of-the-art CNN architectures such as Inception-V4, EfficientNet, and EfficientNetV2 which are a combination not typically explored in diagnostics studies, which often focus solely on either CNNs or transformers. Unlike previous medical classification works that typically focus on binary classification tasks, identifying only the presence or absence of a condition, this study targets a more complex multi-class classification of degenerative lumbar spine conditions by severity levels (normal, moderate, or severe). This approach is critical for clinical decision-making, as it allows prioritization based on case severity. Additionally, beyond accuracy, this research evaluates models on practical metrics including model size, computational efficiency, and training time to assess their feasibility for real-time clinical applications. By incorporating these considerations, the study provides a more comprehensive evaluation than prior studies, which tend to focus predominantly on accuracy without examining real-world applicability.

II. LITERATURE REVIEW

A. *Previous Works in Medical Image Classification Using CNN Based Model*

In medical imaging applications, CNN-based models has been widely used, as demonstrated in a 2022 study [8], which focused on breast cancer detection in mammogram images. This study introduced a powerful hybrid model, combining a Convolutional Neural Network (CNN) with the Inception-V4 architecture CNN-Inception-V4. This innovative model achieved 99.2% accuracy, 99.8% sensitivity, 96.3% specificity, and 97% F1-score, outperforming models like CNN, Inception-V4, and ResNet-50. For which proven its robustness and reliability for tasks in medical image classification. However, the model's design increases computational load, posing challenges for deployment on resource-constrained devices like clinical edge devices.

Another study in 2023 [9] leveraged the EfficientNetV2 model to classify dermatologic manifestations of cardiovascular diseases into seven distinct categories. The model was trained on a dataset of 2,665 images, including categories like cyanosis, liverdo reticularis, and xanthoma, with data augmentation to expand training samples. By using transfer learning, EfficientNetV2 achieved 96.04% accuracy, showing its strength in medical image analysis. While it excelled in capturing colour and texture features, some misclassifications persisted due to class imbalance and limited dataset size.

In 2020, [10] introduced a CNN-based medical decision support system using the EfficientNet architecture for COVID-19 detection in CT and X-ray images. EfficientNet outperformed models such as DarkCovidNET, VGG19, MobileNet v2, CapsNet, and nCOVnet, achieving an accuracy of 99.62% in binary classification (COVID-19 vs. normal) and 96.70% in multiclass classification (COVID-19, pneumonia, and normal). The system demonstrated high precision and recall, with an F1-score of 99.62% for binary and 97.11% for multiclass classification. Using 10-fold stratified cross-validation, EfficientNet proved to be a valuable tool in automated diagnostics, supporting timely and accurate COVID-19 detection for healthcare professionals.

B. *Previous Works in Medical Image Classification Using Transformer Based Model*

In 2023, [11] proposed using Transformer for COVID19 medical image classification. The dataset used was 2 dataset of binary class lung CT scans and X-Ray that's separated to normal, infected virus, infected bacteria, and infected COVID19. The proposed model was a hybrid of ResNet and ViT, named RMT-Net. The proposed model resulted in a validation accuracy of 98.84% on the X-Ray dataset and 99.24% on the CT scans dataset, which outperforms ResNet-50, VGGNet-16, i-CapsNet and MGMADS-3.

Other study in 2024 also uses the ViT architecture [12], it proposes a model codenames CTBViT to detect the presence of Tuberculosis in CT and X-Ray scans. The proposed model differs from the base ViT structure by introducing a patch-reduction-block (PRB) to improve the efficiency and training speed of the model, and a randomized classifier to avoid overfitting. The dataset used was a private Tuberculosis dataset from Fourth People's Hospital of Huai'an City, Jiangsu Province, China. The dataset was separated to two classes which are DS (Drug Sensitive) Tuberculosis and MDR (Drug Resistant) Tuberculosis. The resulting model could accurately classify the presence of DS-TB and MDR-TB with precision of 97.6% and 99%, cumulative accuracy of 98.3%.

C. *Degenerative Lumbar Spine*

Degenerative lumbar spine disease is a significant cause of disability worldwide. This disease includes various conditions such as spondylolisthesis, disc degeneration, and lumbar spinal stenosis. Patients often experience symptoms such as lower back pain (LBP), weakness, and lower extremity pain [27]. The degeneration process is categorized into three stages which are A, B, and C based on the progression's location and sequence [28].

In Stage A of spinal degeneration, the nucleus pulposus that normally functions hydrostatically by evenly transmitting pressure to the surrounding annulus fibrosus and endplates will dehydrate, lowering the intradiscal pressure and increasing the load on the annulus fibrosus. This can lead to cracks, cavities, clefts, and fissures, along with phenomena like the vacuum effect, fluid accumulation, and calcification. The annulus fibrosus, endplates, and bone marrow undergo significant changes in stage B. Fissures form in the annulus and disk material may displace beyond the intervertebral space which will cause bulging or herniation in various directions. Endplate damage is a key marker that is often accompanied by bone marrow changes like edema, fibrous tissue, yellow marrow, or dense bone, each affecting symptoms and spinal stability. These alterations may cause intravertebral instability (both vertical and horizontal) by impacting discs, facet joints, and ligaments, potentially resulting in misalignment and conditions such as degenerative or cervical spondylolisthesis and spondylosis. Stage C impacts facet joints, ligamentum flavum, and the spinal canal. Facet joints may develop osteoarthritis, which can occur independently or due to disc degeneration and narrowing. Ligamentum flavum thickening commonly accompanies disc degeneration and herniation, while spinal canal stenosis may occur, potentially leading to severe cord compression and nerve root changes [28].

D. *Transformer*

The Transformer architecture was first introduced in 2017 [13]. It was developed with only using attention mechanisms to capture the global features between input and output. The architecture of the base transformer itself, rely on a stack of encoder and decoder block. The encoder stack consists of $N = 6$ identical encoder layers, with each layer consist of a multi-head self-attention mechanism and a position-wise feed-forward network, with a residual connection around each of the sub layer and a bath normalization after each sub layer. The output of each sub layer will be $LayerNorm(x + Sublayer(x))$. With each

$Sublayer(x)$ are function for multi-head self-attention mechanism and a position-wise feed-forward network. Resulting in an output dimension of $dmodel$ = 512.

The decoder stack also consists of N = 6 identical decoder layer, with each layer consisting of the same 2 sub layers as the encoder stack. However, each decoder layer includes an additional multi-head attention mechanism that attends to the encoder's outputs. The self-attention layer of the decoder stack is also modified by preventing the position from attending to subsequent positions, with offsetting the output embedding by 1 position. This is done to ensures that the prediction of position i is only predicted by positions less than i. The novel approach of only relying on attention mechanisms, resulted in the base Transformer model establishing the new state-of-the-art $BLEU$ score of 28.4 on WMT 2014 English-to-German translation task [13].

The success of Transformers has revolutionized natural language processing (NLP) and has expanded to computer vision, with models like the Vision Transformer (ViT) achieving state-of-the-art performance on image classification through a fully attention-based framework [14].

E. Convolutional Neural Network (CNN)

Convolutional Neural Network (CNN) is an advanced deep learning architecture, that has been specialized to handle multi-dimensional data structures effectively and automatically (e.g image) [15]. CNN has been a revolution in computer vision since 2012, It was first introduced in its participation of ImageNet Large Scale Visual Recognition Challenge (ILSVRC). CNN is also the current state-of-the-art architecture in many computers vision task [16].

CNN is comprised of neurons that is separated to 3 dimensions, the height, width, and depth. For example, if given an image of size 64×64×3, the output will be 1×1×n (n being the number of class). The overall architecture of CNN itself, consist of Convolution, Pooling, and Fully Connected Layers [17]. CNN distinguishes itself by effectively extracting local features, but it also has a difficulty in capturing global features and long-range tendencies [18].

F. Models

1) Inception-V4

Inception-V4 was introduced in 2016, it builds upon the Inception deep convolutional architecture, which was initially introduced in 2014 codenamed Inception-V1 [19]. Inception-V1 introduced the concept of using multiple filter sizes in parallel, enabling the network to capture various visual features at different scales [26]. Inception-V4 incorporates three types of Inception blocks which are A, B, and C each carefully optimized for different operations to handle various feature extraction tasks effectively. Additionally, it includes reduction blocks specifically designed to adjust the width and height of feature maps, enhancing the network's flexibility in capturing complex patterns. The architecture also expands filter banks to alleviate representation bottlenecks, which improves the model’s learning capacity and efficiency [19].

2) EfficientNet

In 2020, [20] introduced EfficientNet, a series of convolutional neural network models that significantly advanced the field of image classification. The largest model, EfficientNet-B7, achieved a remarkable 84.3% top-1 accuracy, establishing a new state-of-the-art performance at the time while being 8.4 times smaller and 6.4 times faster than GPipe, the previous benchmark model. The exceptional performance of EfficientNet was realized through a technique known as neural architecture search. This approach allowed for the development of a baseline model optimized for floating point operations per second (FLOPS). EfficientNet-B0, the resulting baseline model, consists of convolutional blocks with the mobile inverted bottleneck (MBConv) serving as its primary building block. This makes even the smallest variant, EfficientNet-B0, outperformed models with 4.9 times more parameters and was 11 times faster in execution.

3) EfficientNetV2

EfficientNetV2 was first introduced in 2021. EfficientNetV2 outperforms other models significantly, with using less parameters and faster training. The increase was due to its approach that used training-aware Neural Architecture Search (NAS) and scaling. The architecture of EfficientNetV2 was based upon the original EfficientNet design with several key modifications. EfficientNetV2 extensively utilizes MBConv and fused-MBConv blocks inspired by EfficientNet-EdgeTPU in its early layers. Additionally, it applies smaller expansion ratios within the MBConv layers and reduces kernel sizes to 3×3, while compensating with an increased number of layers. Lastly, EfficientNetV2 omits the final stride-1 stage present in the base EfficientNet model, enhancing efficiency and performance. With larger model variations, reinforcement learning, and random search methods were employed to explore optimal configurations while maintaining computational efficiency [21].

4) Vision Transformer (ViT)

Vision Transformer or ViT was introduced in 2021, which largely contributed to by the success of Transformer model in natural language processing (NLP) tasks. ViT works by treating images like a sentence or documents. It works by splitting an image to patches, with the sequence of linear embedding of the patches as the input for the transformer. Architecturally ViT strongly followed the original Transformer model, however, it only uses the encoder stack. The resulting models are ViT-Base, ViT-Large, and ViT-Huge with the difference being the number of heads in each model of 12, 16, and 16 respectively and number of layers of 12, 24, and 32 respectively. This approach resulted in the ViT-L/16 (ViT-L with patch size of [16 X 16]) outperforming the ResNet152x4, with also being considerably faster [22].

III. Methodology

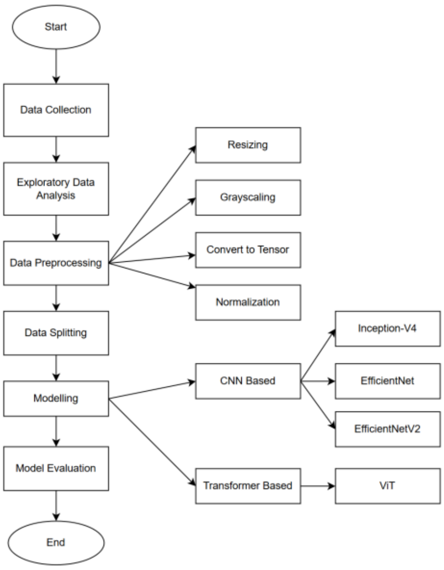

Fig. 1. Research Flow

A. *Data Collection*

The dataset used in this study, titled "RSNA 2024 Lumbar Spine Degenerative Classification," was sourced from the official Kaggle platform. Curated and annotated by the Radiological Society of North America (RSNA) and the American Society of Neuroradiology (ASNR), it contains MRI scans labelled with severity levels (Normal/Mild, Moderate, or Severe) for five degenerative lumbar spine conditions across intervertebral disc levels from L1/L2 to L5/S1. The conditions include Left Neural Foraminal Narrowing, Right Neural Foraminal Narrowing, Left Subarticular Stenosis, Right Subarticular Stenosis, and Spinal Canal Stenosis [23].

B. *Exploratory Data Analysis*

The dataset that has been collected consists of 48,617 data comprises five primary categories relating to spinal abnormalities, each labelled according to severity: "Normal/Mild," "Moderate," and "Severe."

TABLE I. Dataset Distribution

Conditions	**Total Data**	**Data Count**		
		Normal/ Mild	**Moderate**	**Severe**
Spinal Canal Stenosis	9,753	8,552	732	469
Left Neural Foraminal Narrowing	9,860	7,671	1,792	397
Right Neural Foraminal Narrowing	9,829	7,684	1,767	378
Left Subarticular Stenosis	9,603	6,857	1,834	912
Right Subarticular Stenosis	9,612	6,862	1,825	925

Based on the information on TABLE I, "Normal/Mild" samples dominate, while "Severe" cases are markedly underrepresented. This class imbalance is a critical consideration, as it can impact model performance, potentially leading to biased predictions that may overlook "Severe" cases.

C. *Data Preprocessing*

Dataset that has been previously acquired cannot be directly employed for training and testing models, it must first undergo a preprocessing stage. Data preprocessing is an essential process involves preparing raw data and transforming it into a format suitable for the intended task. Preprocessing step itself is needed to eliminate noise/variation, as well as improving the quality and contrast of the image which can affect the accuracy of the model itself [24]. The data preprocessing steps taken in this study are as follows:

1. Resizing the image into the dimension of 128×128 pixels
2. Grayscaling the image to reduce the colour information to a single channel, which simplifies the data.
3. Convert the data to a PyTorch tensor. This conversion not only transforms the data into a format compatible with PyTorch but also rescales the pixel values from the original range of [0, 255] (for uint8 images) to a normalized range of [0.0, 1.0] (for float tensors).
4. Normalize the pixel values using specific mean and standard deviation values to ensure that the input data is consistent and comparable to the data the model was originally trained on.

D. *Data Splitting*

After preprocessing, the training data will be split into a 70:30 ratio, allocating 70% for training and 30% for validation. The validation set is crucial for these complex models to fine-tune hyperparameters, prevent overfitting, and ensure better generalization [13]. Regarding the test data, it has been provided by RSNA, ensuring a standardized evaluation of the model's performance.

E. *Modelling*

During the modelling phase, two types of models are trained: CNN-based models, which include architectures such as Inception-V4, EfficientNet, and EfficientNetV2, and lightweight transformer models, specifically utilizing the Vision Transformer (ViT) architecture. This dual approach allows for leveraging the strengths of both convolutional neural networks and transformer models to enhance performance in the classification of degenerative lumbar spine conditions.

F. *Model Evaluation*

After obtaining results from the testing stage, it is essential to evaluate the effectiveness of each model. Accuracy will serve as the primary metric for this evaluation process. To calculate accuracy, we need to determine the values of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) [25]. Here, TP represents the number of data points correctly classified as positive, FP indicates the number of data points incorrectly classified as positive, TN refers to the number of data points accurately classified as negative, and FN denotes the number of data points inaccurately classified as negative.

Accuracy is an evaluation metric that assesses how well a classification model correctly predicts cases in a dataset. This metric indicates the model's overall ability to make accurate predictions. The following is the formula for calculating the accuracy score [25]:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

IV. EXPERIMENTS AND RESULT

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	Table column subhead	***Subhead***	***Subhead***
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[a.] Sample of a Table footnote. (*Table footnote*)

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V. CONCLUSION (HEADING 5)

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